

5	(a)	Define <i>electric potential</i> at a point. 	[2]
	(a)	Solution: The electric potential V at a point in an electric field is defined as the <u>work done per unit positive charge</u> by an external agent in moving a <u>point charge</u> from infinity to that point (without any change in kinetic energy).	1 1

(b) An isolated solid metal sphere is positively charged.

The variation of the electric potential V with distance x from the centre of the sphere is shown in Fig. 5.1.

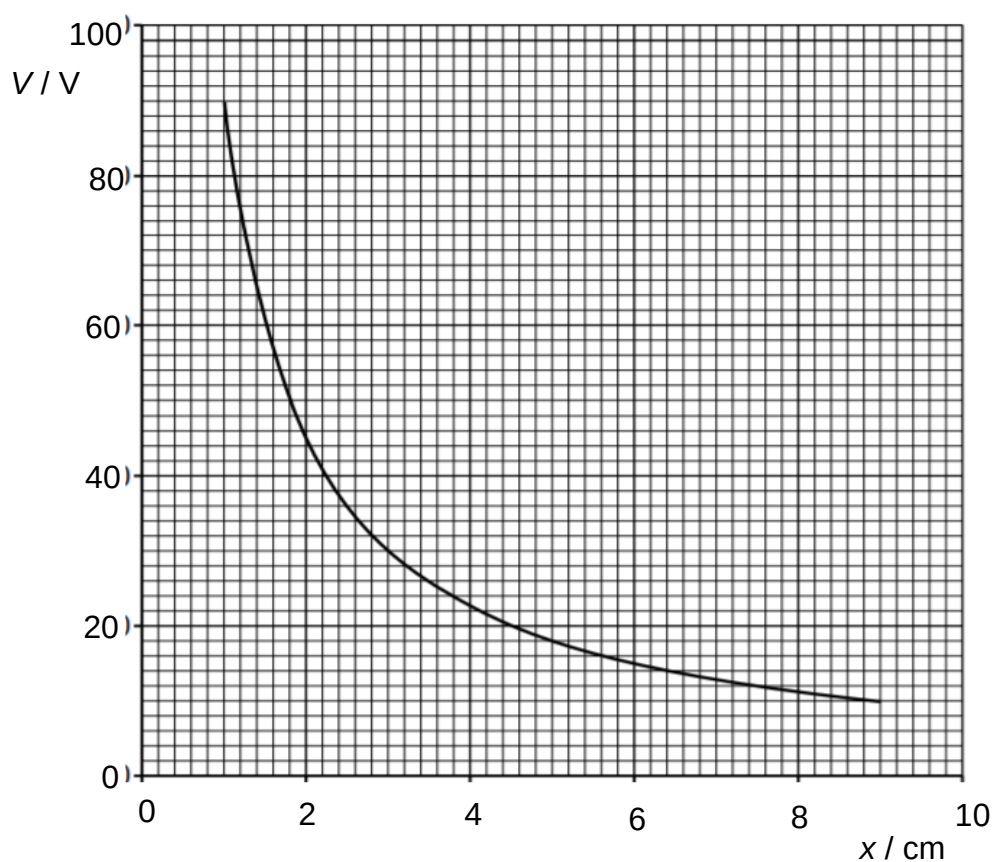


Fig. 5.1

Use Fig. 5.1 to suggest

(i) why the radius of sphere cannot be greater than 1.0 cm,

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[1]

(i) Solution:
 The value of potentials inside the sphere would be constant. So the radius cannot be greater than 1.0 cm.
OR
 The values of potential outside the sphere (for $x > 1.0 \text{ cm}$) is not constant. So the radius cannot be greater than 1.0 cm.

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(ii) quantitatively that the charge on sphere behaves as if it were a point charge.

				[3]																																																
		(ii)	Solution: For point charge, the product Vx is constant because $V = \frac{Q}{4\pi\epsilon_0 x} \Rightarrow Q = 4\pi\epsilon_0 Vx \Rightarrow Q = kVx$ Choose co-ordinates that are clear and determines two values (minimally) of Vx at least 4 cm apart. (or not too close to each other) Since the product of Vx is a constant (or the values of the products are close to each other), the charge on the sphere acts as if it was a point charge. <table><tr><th>V / V</th><th>x / m</th><th>Vx / Vm</th></tr><tr><td>90</td><td>0.010</td><td>0.900</td></tr><tr><td>76</td><td>0.012</td><td>0.912</td></tr><tr><td>66</td><td>0.014</td><td>0.924</td></tr><tr><td>58</td><td>0.016</td><td>0.928</td></tr><tr><td>50</td><td>0.018</td><td>0.900</td></tr><tr><td>45</td><td>0.020</td><td>0.900</td></tr><tr><td>32</td><td>0.028</td><td>0.896</td></tr><tr><td>30</td><td>0.030</td><td>0.900</td></tr><tr><td>28</td><td>0.032</td><td>0.896</td></tr><tr><td>24</td><td>0.038</td><td>0.912</td></tr><tr><td>18</td><td>0.050</td><td>0.900</td></tr><tr><td>16</td><td>0.056</td><td>0.896</td></tr><tr><td>14</td><td>0.064</td><td>0.896</td></tr><tr><td>12</td><td>0.076</td><td>0.912</td></tr><tr><td>10</td><td>0.09</td><td>0.900</td></tr></table>	V / V	x / m	Vx / Vm	90	0.010	0.900	76	0.012	0.912	66	0.014	0.924	58	0.016	0.928	50	0.018	0.900	45	0.020	0.900	32	0.028	0.896	30	0.030	0.900	28	0.032	0.896	24	0.038	0.912	18	0.050	0.900	16	0.056	0.896	14	0.064	0.896	12	0.076	0.912	10	0.09	0.900	1 1 1
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	(c)	An alpha particle of charge $+2e$ is directed to the sphere from infinity with a kinetic energy E_k. Use Fig. 5.1 to determine the minimum value of E_k of the alpha particle to reach a point just 5 cm from the centre of the sphere. $E_k = \dots\dots\dots$ J	[2]
	(c)	Solution: At $x = 5.0$ cm, $V = 18$ V $E_k = q(V)$ $= 2(1.6 \times 10^{-19})(18)$ $= 5.76 \times 10^{-18}$ J or 5.8×10^{-18} J OR By conservation of energy, $E_k + E.P.E_\infty = K.E_p + E.P.E_p$ $E_k + 0 = 0 + qV_p$ $E_k = 2(1.6 \times 10^{-19})(18)$ $= 5.76 \times 10^{-18}$ J or 5.8×10^{-18} J	1 1 1 1

6	(a)	A uniform wire AB of length 150 cm and radius 0.45 mm is connected in series	
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